

MEDTRACK_DV

Dashboard Testing & Validation Report

Hospital Operations & Patient Analytics Dashboard

Quality Assurance, Functional Testing & Dashboard Validation

Hospital Analytics Dashboard Project

1. Executive Summary

MedTrack_DV is a healthcare analytics project delivering an interactive Hospital Operations & Patient Analytics Dashboard suite, built from hospital operational and patient admission data. The purpose of the project is to give hospital stakeholders a consolidated, interactive view of admissions, patient flow, department performance, and resource utilization to support operational decision-making.

As part of Milestone 4 — Module 7: Testing & Validation, structured testing was carried out across the full dashboard suite. Testing objectives covered KPI accuracy, healthcare metric correctness, dashboard interactivity, filter behavior, cross-dashboard navigation, patient-flow analytics, department analytics, resource-utilization metrics, and visual consistency.

All four dashboards — Hospital Overview, Patient Flow, Department Analytics, and Resource Utilization — were tested. Across 28 documented test cases, the suite achieved a 100% test pass rate, with all validated KPIs matching source values within expected display-rounding tolerance — exceeding the required 95% KPI accuracy threshold. No major dashboard issues were identified.

OVERALL RESULT: PASS — 100% Test Pass Rate

2. Project Overview

MedTrack_DV provides a four-dashboard analytics suite for hospital operations and patient care monitoring. The suite is designed to support:

- **Hospital performance monitoring** — admissions, occupancy, revenue, and length-of-stay trends.
- **Patient flow analysis** — admission and discharge trends, peak patient load, and patient movement across wards.
- **Department performance** — patient volumes, readmission rates, and revenue by department.
- **Resource utilization** — bed, doctor, and nurse allocation across departments and ward types.

3. Testing Objectives

Testing for Milestone 4 was conducted to verify the following areas of the dashboard suite:

- KPI accuracy against validated source values
- Healthcare metric correctness (readmission rate, occupancy rate, length of stay, admissions per bed)
- Dashboard functionality across all four dashboards
- Filter functionality (Admission Date, Department, Ward Type, Admission Type)
- Cross-dashboard navigation
- Patient-flow analytics
- Department analytics
- Resource utilization analytics
- Visual consistency and layout integrity

4. KPI Validation

The table below compares each core KPI's validated source value against the value displayed on the dashboard. All KPIs matched within expected rounding tolerance.

KPI	Expected / Source Value	Dashboard Value	Result
Total Admissions	10,000	10K	PASS
Total Patients	4,345	4.345K	PASS
Average Age	53.60	53.60	PASS
Average LOS	6.96 days	6.96	PASS
Readmission Rate	14.91%	15%	PASS
Occupancy Rate	19.64%	19.64%	PASS
Admissions per Bed	21.28	21.28	PASS
Total Departments	7	7	PASS
Total Beds	470	470	PASS
Total Doctors	150	150	PASS
Total Nurses	278	278	PASS

Note: Minor differences such as 14.91% vs. 15%, and 10,000 vs. 10K, are the result of dashboard display rounding and are not considered defects.

5. Dashboard Testing

5.1 Hospital Overview Testing

The following elements of the Hospital Overview dashboard were tested:

- Total Admissions, Readmission Rate, Average Length of Stay, Occupancy Rate, and Admissions per Bed KPI cards
- Monthly Admissions Trend and Monthly Revenue Trend charts
- Admissions by Department and Top Departments by Admissions charts
- Patient Distribution by Ward Type and Average Length of Stay by Department charts
- Admission Date, Department, Ward Type, and Admission Type filters



Figure 1 — Hospital Overview Dashboard

Hospital Overview Dashboard: PASS

5.2 Patient Flow Testing

The following elements of the Patient Flow dashboard were tested:

- Total Patients, Average Age, Readmission Rate, and Average Length of Stay KPI cards
- Patient Movement Analysis and Peak Patient Load Monitoring charts
- Monthly Patient Admissions Trend and Monthly Discharges Trend charts
- Patient Distribution by Gender, Age Group, and Admission Type charts
- Admission Date, Department, Admission Type, and Ward Type filters



Figure 2 — Patient Flow Dashboard

Patient Flow Dashboard: PASS

5.3 Department Analytics Testing

The following elements of the Department Analytics dashboard were tested:

- Total Departments and Readmission Rate KPI cards
- Patients by Department and Readmission Rate by Department charts
- Admissions per Bed by Department and Average Length of Stay by Department charts
- Revenue by Department chart
- Admission Date, Department, Admission Type, and Ward Type filters



Figure 3 — Department Analytics Dashboard

Department Analytics Dashboard: PASS

5.4 Resource Utilization Testing

The following elements of the Resource Utilization dashboard were tested:

- Occupancy Rate, Admissions per Bed, Total Hospital Beds, Total Doctors, and Total Nurses KPI cards
- Beds by Department and Nurses by Department charts
- Staff Allocation by Department and Doctors by Department charts
- Resource Availability by Ward Type chart
- Admission Date, Department, Admission Type, and Ward Type filters

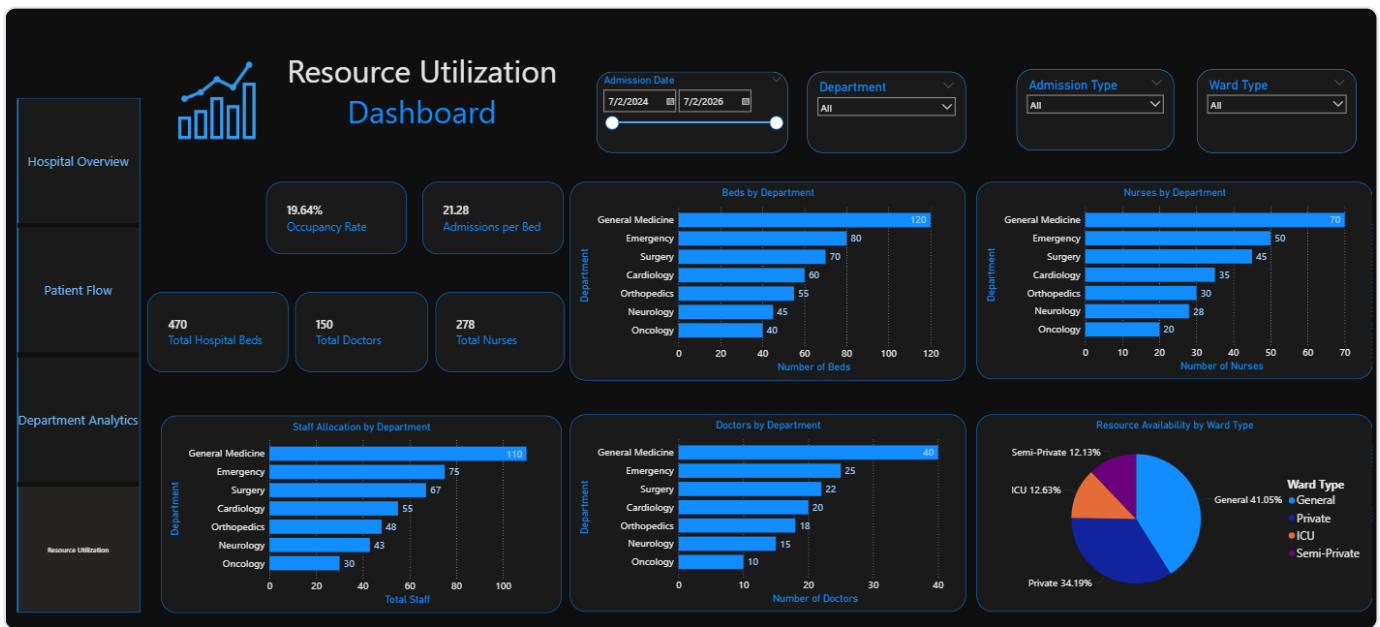


Figure 4 — Resource Utilization Dashboard

Resource Utilization Dashboard: PASS

6. Filter Testing

Filter behavior was verified across all four dashboards.

Filter	Dashboard	Expected Behavior	Result
Admission Date	All dashboards	Updates dashboard data for selected date range	PASS
Department	All dashboards	Filters visuals to selected department	PASS
Ward Type	All dashboards	Filters visuals to selected ward category	PASS
Admission Type	All dashboards	Filters visuals to selected admission category	PASS

7. Navigation Testing

Navigation between all four dashboards was tested and confirmed functional in both directions of the review flow:

- Hospital Overview → Patient Flow
- Patient Flow → Department Analytics
- Department Analytics → Resource Utilization
- Resource Utilization → Hospital Overview

Navigation Testing: PASS

8. Visual Quality Testing

Visual quality checks were performed across the full dashboard suite, confirming:

- Consistent dark healthcare theme with blue accent color throughout
- Consistent typography and chart styling across all dashboards
- Clear, readable KPI cards and proper element alignment
- No major overlapping elements or broken visualizations
- Readable chart titles and KPI labels
- Functional filters and a consistently visible navigation panel

Visual Quality Testing: PASS

9. Defect Summary

Severity	Count
Critical	0
High	0
Medium	0
Low	0

Conclusion: No major dashboard issues were identified during testing.

10. Final Testing Summary

Metric	Result
Total Test Cases	28
Passed	28 (100%)
Failed	0
Pending	0
Overall Status	PASSED

The MedTrack_DV dashboard suite successfully passed functional, KPI, interaction, patient-flow, department, resource-utilization, navigation, and visual validation testing.

11. Final Conclusion

Based on the results of this testing cycle, the MedTrack_DV dashboard suite is ready for final project delivery and portfolio presentation. The dashboard satisfies the Milestone 4 testing requirements defined in Module 7, with KPI accuracy and dashboard functionality both meeting the project's evaluation criteria.

"All four dashboards — Hospital Overview, Patient Flow, Department Analytics, and Resource Utilization — passed testing with a 100% pass rate and KPI accuracy exceeding the 95% target."

OVERALL STATUS: PASSED